1. Wireless Security

1. What is a wireless network?

A wireless network connects devices using radio waves instead of physical cables.

2. What is an SSID?

SSID (Service Set Identifier) is the name used to identify a Wi-Fi network.

3. Why do wireless networks require encryption?

Encryption protects transmitted data from unauthorized access and eavesdropping.

4. Why is WEP considered insecure?

WEP is insecure because it uses weak encryption that can be cracked relatively easily.

5. What improvements did WPA2 and WPA3 introduce?

WPA2 introduced stronger encryption using AES-CCMP. WPA3 improves security with SAE authentication, better protection against password-guessing attacks, and stronger security options.

2. Wireless Assessment

6. What is Aircrack-ng?

Aircrack-ng is an open-source suite of tools used to assess Wi-Fi network security.

7. Why do security professionals use wireless assessment tools?

To identify vulnerabilities, analyze wireless traffic, and evaluate Wi-Fi security.

8. What is the purpose of packet analysis?

Packet analysis examines network packets to understand communication, troubleshoot problems, and detect suspicious activity.

9. Why must wireless testing always be authorized?

To respect privacy, avoid disrupting networks, and ensure testing is legally permitted.

3. Social Engineering

10. What is social engineering?

Social engineering is the manipulation of people into revealing sensitive information or performing unsafe actions.

11. Why do attackers target people?

Because human emotions and mistakes can help attackers bypass technical security controls.

12. What is pretexting?

Pretexting is a social engineering technique in which an attacker invents a believable story or identity to obtain information or access.

13. What are common warning signs of social engineering attacks?

Urgency, unfamiliar identities, unexpected requests, requests for confidential information, and attempts to bypass security procedures.

4. Phishing

14. What is phishing?

Phishing is a cyberattack in which attackers use fraudulent emails or messages to steal information or trick users into unsafe actions.

15. What are common phishing indicators?

Suspicious sender addresses, urgent language, unexpected links, suspicious attachments, and requests for sensitive information.

16. Why should users verify unexpected requests?

To confirm their legitimacy and prevent unauthorized access, fraud, and data theft.

17. Why are phishing simulations useful?

They help users recognize phishing attempts, improve security awareness, and practice reporting suspicious messages in a controlled environment.